2-d Kernels in Quotients of the Square Grid and in Three-Dimensional Tori with a Triangle Factor

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Abstract

A 2-d kernel of a graph is an independent set S such that every vertex outside S has at least two neighbours in S. We determine all 2-d kernels of every finite simple 4-regular graph obtained from the square grid $\mathbb{Z}^2$ as a quotient by a lattice. Every such kernel is the image of a *diagonal* set $\{(x,y) : x - sy \in K\}$ with $s = \pm 1$, and their number is $P(d_+) + P(d_-) - 2\,[\,d_+, d_-$ both even$\,]$, where P is the Perrin sequence and $d_\pm$ are two explicit periods of the lattice. Three corollaries follow. The torus $C_m \,\square\, C_n$ $(m, n \geq 3)$ has a 2-d kernel if and only if $\gcd(m,n) > 1$, and it then has exactly $2P(d) - 2\,[\,d$ even$\,]$ of them, where $d = \gcd(m,n)$; this corrects Theorem 2.5 and Corollary 2.6 of Bednarz and Włoch (2016). The number of 2-d kernels of the tensor product $C_m \times C_n$ is determined for all $m, n \geq 3$, which extends the case $n = 3$ due to Bednarz (2021). The circulant $C_N(a,b)$ has a 2-d kernel if and only if $\gcd(N, a+b) > 1$ or $\gcd(N, a-b) > 1$. Finally, we show that $C_3 \,\square\, C_b \,\square\, C_c$ $(b, c \geq 3)$ has a 2-d kernel if and only if $3 \mid b$ or $3 \mid c$, by reducing to fall 3-colourings of $C_b \,\square\, C_c$, which we classify and count.

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1 Introduction

1.1 Definitions and conventions

All graphs are simple. For a graph G and a vertex v we write $N_G(v)$ for the set of neighbours of v, and we drop the subscript when the graph is clear. For an integer $k \geq 1$ we write $\mathbb{Z}_k = \mathbb{Z}/k\mathbb{Z}$, and for a statement Q the *Iverson bracket* $[\,Q\,]$ is 1 if Q is true and 0 otherwise.

Definition 1.1 (2-d kernel)**.** A set $S \subseteq V(G)$ is a 2-*d kernel* of G if

(K1) S is independent, that is, no two vertices of S are adjacent, and

(K2) every vertex $v \notin S$ satisfies $|N_G(v) \cap S| \geq 2$.

We write $\sigma(G)$ for the number of 2-d kernels of a finite graph G.

The notion was introduced by A. Włoch [11] under the name (2-d)-kernel. It also appears as an *independent* $(1,1)$-*dominating set*; see [3].

Definition 1.2 (Perrin numbers)**.** The Perrin sequence is defined by $P(0) = 3$, $P(1) = 0$, $P(2) = 2$ and $P(k) = P(k-2) + P(k-3)$ for $k \geq 3$. Its first terms are

k	0	1	2	3	4	5	6	7	8	9	10	11	12
$P(k)$	3	0	2	3	2	5	5	7	10	12	17	22	29

Definition 1.3 (cycles and products)**.** For $k \geq 3$ the cycle C_k has vertex set $\mathbb{Z}_k$, and u, v are adjacent if $v - u \in \{1, -1\}$. Let G and H be graphs.

- The *Cartesian product* $G \,\square\, H$ has vertex set $V(G) \times V(H)$, and (g,h) is adjacent to (g',h') if either $g = g'$ and $hh' \in E(H)$, or $h = h'$ and $gg' \in E(G)$.
- The *tensor product* $G \times H$ has vertex set $V(G) \times V(H)$, and (g,h) is adjacent to (g',h') if $gg' \in E(G)$ and $hh' \in E(H)$.

The Cartesian product is associative up to the obvious isomorphism [7], so $C_a \,\square\, C_b \,\square\, C_c$ is the graph on $\mathbb{Z}_a \times \mathbb{Z}_b \times \mathbb{Z}_c$ in which two vertices are adjacent when they differ in exactly one coordinate, and there by ± 1.

1.2 Main results

Theorem 1.4 (tori; Corollary 6.1)**.** *Let $m, n \geq 3$ and $d = \gcd(m, n)$. Then $C_m \,\square\, C_n$ has a 2-d kernel if and only if $d \geq 2$. Every 2-d kernel has the form $\{(x, y) \in \mathbb{Z}_m \times \mathbb{Z}_n : (x - sy) \bmod d \in \overline{K}\}$ with $s \in \{+1, -1\}$ and $\overline{K}$ a maximal independent set of the cycle on $\mathbb{Z}_d$ (for $d = 2$, one of $\{0\}$ and $\{1\}$), and*

$$\sigma(C_m \,\square\, C_n) = 2P(d) - 2\,[\,d \text{ even}\,].$$

The sizes of the 2-d kernels are exactly the numbers mnk/d with $\lceil d/3 \rceil \leq k \leq \lfloor d/2 \rfloor$.

Theorem 1.4 is a special case of a general statement about quotients of the square grid (Theorem 5.6), which also gives the number of 2-d kernels of every tensor product of two cycles (Corollary 6.3) and of every 4-regular circulant satisfying a mild condition (Corollary 6.5). In three dimensions we prove the following.

Theorem 1.5 (Corollary 7.12)**.** *Let $b, c \geq 3$. The graph $C_3 \,\square\, C_b \,\square\, C_c$ has a 2-d kernel if and only if $3 \mid b$ or $3 \mid c$. More precisely, writing $\chi(k) = 2^k + 2(-1)^k$,*

$$\sigma(C_3 \,\square\, C_b \,\square\, C_c) = 2\chi(c)\,[\,3 \mid b\,] + 2\chi(b)\,[\,3 \mid c\,] - 12\,[\,3 \mid b \text{ and } 3 \mid c\,].$$

1.3 Relation to the literature

Bednarz and Włoch [1, Theorem 2.5] state that for $n, m \geq 3$ the Cartesian product $C_n \square C_m$ has a 2-d kernel if and only if n and m are both even or $n = m$, and [1, Corollary 2.6] states that the number of 2-d kernels is 2 when m, n are both even and $2n$ for $C_n \,\square\, C_n$ with n odd. Theorem 1.4 shows that both statements are false. Section 8 gives explicit counterexamples and identifies the two steps of their proof that fail. Bednarz [2] studied the tensor product and proved $\sigma(C_n \times C_3) = 3 + P(n)$; Corollary 6.3 gives $\sigma(C_m \times C_n)$ for all $m, n \geq 3$ and contains that result as the case $m = 3$.

Fall colourings (Definition 7.1) were introduced by Dunbar et al. [5]. Kaul and Mitillos [9] proved that if G has a fall k-colouring and H has a proper k-colouring then $G \,\square\, H$ has a fall k-colouring; for $k = 3$, $G = C_b$ with $3 \mid b$ and $H = C_c$ this gives the "if" part of Theorem 7.10. We prove that part again for completeness. We have not found the "only if" part, the classification, or the count in the literature.

1.4 Method and organisation

Each of the two-dimensional graphs above is a quotient $\mathbb{Z}^2/L$ of the infinite square grid by a lattice L, and the quotient map is a covering. Coverings preserve 2-d kernels in both directions (Lemma 5.3), so it suffices to classify the 2-d kernels of $\mathbb{Z}^2$ (Theorem 4.9) and then decide which of them are invariant under L (Theorem 5.6). The classification on $\mathbb{Z}^2$ is the heart of the paper. Every row and every column of a 2-d kernel is a maximal independent set of the line (Lemma 4.2), and the way consecutive rows interlock forces every row to be the previous row shifted by one and the same $s = \pm 1$ (Section 4).

Section 2 lists the standard facts we use. Section 3 treats maximal independent sets of the line and of cycles, Section 4 the grid, Section 5 quotients, and Section 6 the three families. Section 7 contains the three-dimensional result, Section 8 the comparison with [1], and Section 9 the computer checks.

2 Standard facts

We use the following standard facts. The first three are in every introduction to number theory, for example [10].

Fact 2.1. *Every subgroup of $\mathbb{Z}$ equals $d\mathbb{Z}$ for exactly one integer $d \geq 0$.*

Fact 2.2 (Bézout)**.** *For integers $a_1, \ldots, a_r$, not all zero, $a_1\mathbb{Z}+\cdots+a_r\mathbb{Z} = g\mathbb{Z}$ with $g = \gcd(a_1, \ldots, a_r)$. In particular there are integers $t_1, \ldots, t_r$ with $t_1a_1 + \cdots + t_ra_r = g$.*

Fact 2.3 (linear congruences)**.** *Let $N \geq 1$ and $\alpha, \beta \in \mathbb{Z}$. The congruence $\alpha y \equiv \beta \pmod N$ has a solution $y \in \mathbb{Z}$ if and only if $\gcd(\alpha, N)$ divides β.*

Fact 2.4 (fibres of a homomorphism)**.** *Let $\theta : A \to B$ be a surjective homomorphism of finite abelian groups. Then every fibre $\theta^{-1}(b)$ is a coset of $\ker\theta$, so it has $|\ker\theta| = |A|/|B|$ elements.*

Fact 2.5 (counting walks)**.** *Let A be the adjacency matrix of a finite digraph D (with $A_{uv} = 1$ if $u \to v$ is an arc and 0 otherwise). For $k \geq 1$ the entry $(A^k)_{uv}$ is the number of sequences $(w_0, w_1, \ldots, w_k)$ of vertices with $w_0 = u$, $w_k = v$ and $w_t \to w_{t+1}$ an arc for every t. Indeed, expanding the matrix product, $(A^k)_{uv} = \sum_{w_1,\ldots,w_{k-1}} A_{uw_1}A_{w_1w_2}\cdots A_{w_{k-1}v}$, and each summand is 1 exactly for such a sequence and 0 otherwise. Consequently $\operatorname{tr}(A^k)$ is the number of closed walks of length k with a marked starting vertex.*

Fact 2.6 (Cayley–Hamilton)**.** *Every square matrix satisfies its characteristic polynomial; see, for example, [8].*

Fact 2.7 (maximal independent sets)**.** *An independent set I of a graph is maximal (with respect to inclusion) if and only if every vertex outside I has a neighbour in I. Indeed, $I \cup \{v\}$ is independent exactly when v has no neighbour in I.*

3 Maximal independent sets of the line and of cycles

We regard $\mathbb{Z}$ as the vertex set of the two-way infinite path, in which x and $x+1$ are adjacent.

Definition 3.1. A set $K \subseteq \mathbb{Z}$ is an *MIS of* $\mathbb{Z}$ if

(M1) there is no $x \in \mathbb{Z}$ with $x \in K$ and $x+1 \in K$, and

(M2) there is no $x \in \mathbb{Z}$ with $x \notin K$, $x+1 \notin K$ and $x+2 \notin K$.

By Fact 2.7, these are exactly the maximal independent sets of the path on $\mathbb{Z}$: (M1) is independence, and (M2) says that every $x+1 \notin K$ has a neighbour x or $x+2$ in K.

3.1 Enumerations and gaps

Lemma 3.2 (enumerations)**.** *Let $K \subseteq \mathbb{Z}$ be unbounded above and below. Then there is a strictly increasing sequence $(p_i)_{i\in\mathbb{Z}}$ with $K = \{p_i : i \in \mathbb{Z}\}$, and for every such sequence:*

(a) no element of K lies strictly between p_i and p_{i+1};

(b) every integer x satisfies $p_i \leq x < p_{i+1}$ for exactly one i, so $\mathbb{Z}$ is the disjoint union of the sets $\{p_i\}$ and the open intervals (p_i, p_{i+1});

(c) if $(p'_i)_{i\in\mathbb{Z}}$ is another strictly increasing sequence with $K = \{p'_i : i \in \mathbb{Z}\}$, then there is $c \in \mathbb{Z}$ with $p'_i = p_{i+c}$ for all i.

We call such a sequence an enumeration *of K.*

Proof. Existence. K is nonempty because it is unbounded; choose $p_0 \in K$. For $i \geq 0$ put $p_{i+1} = \min\{k \in K : k > p_i\}$; the set is nonempty because K is unbounded above, and it is bounded below by p_i, so the minimum exists. For $i \leq 0$ put $p_{i-1} = \max\{k \in K : k < p_i\}$, which exists for the symmetric reason. The sequence is strictly increasing by construction, and since its terms are integers, $p_i \geq p_0 + i$ for $i \geq 0$ and $p_i \leq p_0 + i$ for $i \leq 0$; so it is unbounded above and below.

(a) For the constructed sequence this holds by the choice of p_{i+1} as a minimum (for $i \geq 0$) or of p_i as a maximum (for $i < 0$). For an arbitrary strictly increasing sequence with $K = \{p_j\}$: an element of K strictly between p_i and p_{i+1} would be some p_j with $p_i < p_j < p_{i+1}$, so $i < j < i + 1$, which is impossible.

(b) Let $x \in \mathbb{Z}$. The set $\{i : p_i \leq x\}$ is nonempty (the sequence is unbounded below) and bounded above (it is unbounded above and increasing), so it has a largest element i. Then $p_i \leq x < p_{i+1}$. This i is unique because the intervals $[p_i, p_{i+1})$ are pairwise disjoint.

$K = \{p_i\}$ for the constructed sequence. Each $p_i \in K$. Conversely, if $x \in K$, take i with $p_i \leq x < p_{i+1}$ by (b); by (a), x is not strictly between p_i and p_{i+1}, so $x = p_i$.

(c) Since $p'_0 \in K$, there is c with $p'_0 = p_c$. For any strictly increasing enumeration (q_j) of K, (a) and (b) show that q_{j+1} is the smallest element of K larger than q_j, and q_{j-1} is the largest element of K smaller than q_j. Applying this to both sequences, induction on $i \geq 0$ gives $p'_i = p_{i+c}$, and induction on $i \leq 0$ gives the same. □

Lemma 3.3 (gaps)**.** *Let K be an MIS of $\mathbb{Z}$. Then K is unbounded above and below, and for every enumeration (p_i) of K the* gaps *$g_i = p_{i+1} - p_i$ satisfy $g_i \in \{2, 3\}$. Conversely, if K is unbounded above and below and some enumeration has all gaps in $\{2, 3\}$, then K is an MIS of $\mathbb{Z}$.*

Proof. Suppose K were bounded above by b. Then $b + 1, b + 2, b + 3 \notin K$, contrary to (M2). The case of a lower bound is symmetric. So an enumeration exists by Lemma 3.2.

- $g_i \neq 1$: otherwise p_i and $p_i + 1$ both lie in K, contrary to (M1).
- $g_i \leq 3$: otherwise $p_i + 1$, $p_i + 2$, $p_i + 3$ are all strictly between p_i and p_{i+1}, hence not in K by Lemma 3.2(a), contrary to (M2).

Since $g_i \geq 1$ (the sequence is strictly increasing), $g_i \in \{2, 3\}$.

Conversely, assume all gaps are 2 or 3. If $x, x + 1 \in K$, then $x = p_i$ for some i and $x + 1 = p_j$ with $j > i$, so $g_i \leq p_j - p_i = 1$, a contradiction; so (M1) holds. For (M2), let $x \in \mathbb{Z}$; we show that one of $x, x + 1, x + 2$ lies in K. Take i with $p_i \leq x < p_{i+1}$ (Lemma 3.2(b)). If $x = p_i$, then $x \in K$. Otherwise $x \geq p_i + 1$, so $p_{i+1} = p_i + g_i \leq p_i + 3 \leq x + 2$; together with $p_{i+1} > x$ this gives $p_{i+1} \in \{x + 1, x + 2\}$, and $p_{i+1} \in K$. □

Lemma 3.4 (one point in each gap)**.** *Let A and B be MIS of $\mathbb{Z}$ with $A \cap B = \emptyset$, and let (p_i) enumerate A, with gaps g_i. Then for every i the open interval (p_i, p_{i+1}) contains exactly one element of B; call it b_i. Moreover*

$$b_i - p_i \in \{1, 2\} \qquad \text{and} \qquad b_i - p_i \leq g_i - 1,$$

and $(b_i)_{i\in\mathbb{Z}}$ is an enumeration of B.

Proof. Fix i and write $g = g_i \in \{2, 3\}$ (Lemma 3.3). The integers strictly between p_i and p_{i+1} are $p_i + 1, \ldots, p_i + g - 1$: one integer if $g = 2$, and the two consecutive integers $p_i + 1, p_i + 2$ if $g = 3$.

- *At most one.* Two elements of B in the interval would be $p_i + 1$ and $p_i + 2$, which are consecutive, contrary to (M1) for B.
- *At least one.* If none of them lies in B, then, because $p_i, p_{i+1} \in A$ and $A \cap B = \emptyset$, all $g+1 \geq 3$ consecutive integers $p_i, p_i + 1, \ldots, p_{i+1}$ avoid B, contrary to (M2) for B.

So the interval contains exactly one element b_i of B, and $1 \leq b_i - p_i \leq g - 1 \leq 2$.

By Lemma 3.2(b) applied to A, every integer is either some $p_j \in A$ or lies in exactly one interval (p_j, p_{j+1}). Elements of B are not in A, so every element of B lies in some interval and therefore equals the corresponding b_j. Hence $B = \{b_i : i \in \mathbb{Z}\}$. Finally $b_i < p_{i+1} < b_{i+1}$, so (b_i) is strictly increasing and is an enumeration of B. □

3.2 Periodic MIS and cycles

Definition 3.5. Let $d \geq 1$. A set $K \subseteq \mathbb{Z}$ is *d-periodic* if $K + d = K$. We write $M(d)$ for the number of d-periodic MIS of $\mathbb{Z}$, and $q_d : \mathbb{Z} \to \mathbb{Z}_d$ for reduction modulo d.

If K is d-periodic, then $K - d = K$ as well (subtract d from both sides of $K + d = K$), and by induction $K + td = K$ for every $t \in \mathbb{Z}$. Hence $x \in K$ depends only on $q_d(x)$, and $K = q_d^{-1}(\overline{K})$ for $\overline{K} = q_d(K) \subseteq \mathbb{Z}_d$. Conversely, for every $\overline{K} \subseteq \mathbb{Z}_d$ the set $q_d^{-1}(\overline{K})$ is d-periodic. So $K \mapsto \overline{K}$ is a bijection from d-periodic subsets of $\mathbb{Z}$ to subsets of $\mathbb{Z}_d$.

Lemma 3.6. *Let $d \geq 3$ and let $K \subseteq \mathbb{Z}$ be d-periodic. Then K is an MIS of $\mathbb{Z}$ if and only if $\overline{K}$ is a maximal independent set of the cycle C_d.*

Proof. Since $K = q_d^{-1}(\overline{K})$ and $q_d(x+1) = q_d(x)+1$, for every $x \in \mathbb{Z}$ we have $x \in K \iff q_d(x) \in \overline{K}$, $x + 1 \in K \iff q_d(x) + 1 \in \overline{K}$, and similarly for $x + 2$. Every element of $\mathbb{Z}_d$ is $q_d(x)$ for some x. Therefore:

- (M1) for K holds if and only if there is no $u \in \mathbb{Z}_d$ with $u, u + 1 \in \overline{K}$. Since $d \geq 3$, the edges of C_d are exactly the pairs $\{u, u + 1\}$, so this is independence of $\overline{K}$ in C_d.
- (M2) for K holds if and only if there is no $u \in \mathbb{Z}_d$ with $u, u + 1, u + 2 \notin \overline{K}$; that is, every vertex $u + 1 \notin \overline{K}$ has one of its two neighbours $u, u + 2$ in $\overline{K}$.

By Fact 2.7, the two conditions together say that $\overline{K}$ is a maximal independent set of C_d. □

Lemma 3.7 (Perrin count)**.** *For every $d \geq 1$, $M(d) = P(d)$. In particular $M(1) = 0$, and $M(2) = 2$ with the two 2-periodic MIS being $2\mathbb{Z}$ and $2\mathbb{Z} + 1$.*

Proof. $d = 1$. The 1-periodic sets are $\emptyset$ and $\mathbb{Z}$. The empty set violates (M2) and $\mathbb{Z}$ violates (M1). So $M(1) = 0 = P(1)$.

$d = 2$. The 2-periodic sets are $\emptyset$, $\mathbb{Z}$, $2\mathbb{Z}$ and $2\mathbb{Z}+1$. The first two fail as before. The sets $2\mathbb{Z}$ and $2\mathbb{Z} + 1$ are unbounded in both directions with all gaps equal to 2, so they are MIS by Lemma 3.3. So $M(2) = 2 = P(2)$.

$d \geq 3$. By Lemma 3.6 and the bijection $K \mapsto \overline{K}$, $M(d)$ is the number of maximal independent sets of C_d. Encode $\overline{K}$ by the word $w : \mathbb{Z}_d \to \{0, 1\}$ with $w_u = 1$ if and only if $u \in \overline{K}$. By the proof of Lemma 3.6, $\overline{K}$ is a maximal independent set if and only if

(W1) there is no $u \in \mathbb{Z}_d$ with $w_u w_{u+1} = 11$, and

(W2) there is no $u \in \mathbb{Z}_d$ with $w_u w_{u+1} w_{u+2} = 000$.

Let $\Sigma = \{00, 01, 10\}$ and let D be the digraph on Σ with an arc $ab \to bc$ whenever $ab \in \Sigma$, $bc \in \Sigma$ and $abc \neq 000$. Checking the possibilities for each state:

- from 00: the candidates are 00 (giving 000, excluded) and 01 (allowed);
- from 01: the candidates are 10 (allowed) and 11 (not in Σ);
- from 10: the candidates are 00 (giving 100, allowed) and 01 (giving 101, allowed).

So the arcs are $00 \to 01$, $01 \to 10$, $10 \to 00$ and $10 \to 01$.

Words and closed walks. For a word w put $\tau_u = w_u w_{u+1}$. If w satisfies (W1) and (W2), then every $\tau_u \in \Sigma$ by (W1), and $\tau_u = w_u w_{u+1} \to w_{u+1} w_{u+2} = \tau_{u+1}$ is an arc by (W2); so $(\tau_0, \tau_1, \ldots, \tau_{d-1})$ is a closed walk of length d in D starting at τ_0. Conversely, let $(\tau_0, \ldots, \tau_{d-1})$ be a closed walk, meaning that $\tau_u \to \tau_{u+1}$ is an arc for every $u \in \mathbb{Z}_d$. Define w_u to be the first letter of τ_u. Every arc has the form $ab \to bc$, so the second letter of τ_u equals the first letter of τ_{u+1}, which is w_{u+1}; hence $\tau_u = w_u w_{u+1}$, and w satisfies (W1) (as $\tau_u \in \Sigma$) and (W2) (as each step is an arc). These two constructions are inverse to each other. By Fact 2.5, $M(d) = \operatorname{tr}(A^d)$, where A is the adjacency matrix of D.

The trace. In the order $(00, 01, 10)$,

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \lambda I - A = \begin{pmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ -1 & -1 & \lambda \end{pmatrix}.$$

Expanding the determinant along the first row,

$$\det(\lambda I - A) = \lambda \det \begin{pmatrix} \lambda & -1 \\ -1 & \lambda \end{pmatrix} - (-1) \det \begin{pmatrix} 0 & -1 \\ -1 & \lambda \end{pmatrix} = \lambda(\lambda^2 - 1) + \big(0 \cdot \lambda - (-1)(-1)\big) = \lambda^3 - \lambda - 1.$$

By Fact 2.6, $A^3 = A + I$. Multiplying by A^{k-3} gives $A^k = A^{k-2} + A^{k-3}$ for $k \geq 3$, and taking traces, $t_k = \operatorname{tr}(A^k)$ satisfies $t_k = t_{k-2} + t_{k-3}$ for $k \geq 3$. The initial values are $t_0 = \operatorname{tr} I = 3$, $t_1 = \operatorname{tr} A = 0$, and, since

$$A^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix},$$

$t_2 = 0 + 1 + 1 = 2$. These are the defining recurrence and initial values of the Perrin sequence, so $t_k = P(k)$ for all $k \geq 0$, and $M(d) = P(d)$. □

Remark 3.8. That the number of maximal independent sets of C_d is the Perrin number $P(d)$ is due to Füredi [6]; we included a proof to keep the paper self-contained.

Lemma 3.9. *$P(d) \geq 2$ for every $d \geq 2$.*

Proof. $P(2) = 2$, $P(3) = 3$ and $P(4) = 2$. For $d \geq 5$ we have $d - 2 \geq 3$ and $d - 3 \geq 2$, so by strong induction $P(d) = P(d-2) + P(d-3) \geq 2 + 2 \geq 2$. □

Lemma 3.10 (sizes)**.** *Let $d \geq 2$. The numbers $|\overline{K}|$, for K a d-periodic MIS of $\mathbb{Z}$, are exactly the integers k with $\lceil d/3 \rceil \leq k \leq \lfloor d/2 \rfloor$.*

Proof. For $d = 2$ the two sets $\overline{K}$ are $\{0\}$ and $\{1\}$, of size 1, and $\lceil 2/3 \rceil = \lfloor 2/2 \rfloor = 1$. Let $d \geq 3$.

Necessity. Let K be a d-periodic MIS with enumeration (p_i), and let $k = |\overline{K}|$. Then $k \geq 1$, since $K \neq \emptyset$. The elements of K in the half-open interval $[p_0, p_0 + d)$ are $p_0, p_1, \ldots, p_{k-1}$, because q_d is a bijection from this interval onto $\mathbb{Z}_d$ and maps $K \cap [p_0, p_0 + d)$ onto $\overline{K}$. The next element of K after p_{k-1} is $p_0 + d$ (it lies in K by periodicity, and nothing in K lies strictly between p_{k-1} and $p_0 + d$), so $p_k = p_0 + d$ and $g_0 + g_1 + \cdots + g_{k-1} = p_k - p_0 = d$. Each gap is 2 or 3 (Lemma 3.3), so $2k \leq d \leq 3k$, which is $\lceil d/3 \rceil \leq k \leq \lfloor d/2 \rfloor$.

Sufficiency. Let k satisfy $2k \leq d \leq 3k$. Put $r = d - 2k$, so $0 \leq r \leq k$. Let $p_0 = 0$, let the gaps $g_0, \ldots, g_{k-1}$ consist of r entries equal to 3 and $k - r$ entries equal to 2, and extend periodically by $g_{i+k} = g_i$. Their sum over one period is $3r + 2(k - r) = 2k + r = d$. The resulting set $K = \{p_i\}$ satisfies $K + d = K$ (since $p_{i+k} = p_i + d$), is unbounded both ways, has all gaps in $\{2, 3\}$, and is therefore an MIS by Lemma 3.3, with $|\overline{K}| = k$. □

4 The square grid

Let Γ denote the infinite square grid: its vertex set is $\mathbb{Z}^2$, and (x, y) is adjacent exactly to the four vertices $(x \pm 1, y)$ and $(x, y \pm 1)$. For $T \subseteq \mathbb{Z}^2$ the *row* at height y and the *column* at abscissa x are

$$T_y = \{x \in \mathbb{Z} : (x, y) \in T\}, \qquad T^x = \{y \in \mathbb{Z} : (x, y) \in T\}.$$

A set $T \subseteq \mathbb{Z}^2$ is a 2-d kernel of Γ if it satisfies (K1) and (K2) of Definition 1.1 in Γ.

Definition 4.1 (diagonal sets)**.** For $s \in \{+1, -1\}$ define $\varphi_s : \mathbb{Z}^2 \to \mathbb{Z}$ by $\varphi_s(x, y) = x - sy$, and for $K \subseteq \mathbb{Z}$ put

$$D_s(K) = \varphi_s^{-1}(K) = \{(x, y) \in \mathbb{Z}^2 : x - sy \in K\}.$$

Since $(x, y) \in D_s(K) \iff x \in K + sy$, the rows are $D_s(K)_y = K + sy$: each row is the row below it shifted by s.

4.1 Rows and columns

Lemma 4.2 (line lemma)**.** *A set $T \subseteq \mathbb{Z}^2$ is a 2-d kernel of Γ if and only if every row T_y and every column T^x is an MIS of $\mathbb{Z}$.*

Proof. ($\Leftarrow$) *Independence.* Every edge of Γ joins (x, y) to $(x + 1, y)$ or to $(x, y + 1)$ for some x, y. Both ends of the first kind of edge in T would put $x, x + 1$ in T_y, and of the second kind would put $y, y + 1$ in T^x; both are excluded by (M1).

Domination. Let $v = (x, y) \notin T$, so $x \notin T_y$ and $y \notin T^x$. By (M2) for T_y applied to $x - 1, x, x + 1$, at least one of $x - 1$ and $x + 1$ lies in T_y; that is, $(x - 1, y) \in T$ or $(x + 1, y) \in T$. In the same way $(x, y - 1) \in T$ or $(x, y + 1) \in T$. The first vertex differs from v in the first coordinate and the second in the second coordinate, so they are two distinct neighbours of v in T.

($\Rightarrow$) *(M1).* If $x, x + 1 \in T_y$ then the adjacent vertices (x, y) and $(x + 1, y)$ lie in T, contrary to (K1); similarly for columns.

(M2) for rows. Suppose that $(x - 1, y)$, (x, y) and $(x + 1, y)$ are not in T.

(i) The neighbours of (x, y) are $(x - 1, y)$, $(x + 1, y)$, $(x, y - 1)$ and $(x, y + 1)$, and the first two are not in T. Since $(x, y) \notin T$, (K2) says that at least two neighbours are in T, so $(x, y - 1) \in T$ and $(x, y + 1) \in T$.

(ii) The neighbours of $(x - 1, y)$ are $(x - 2, y)$, (x, y), $(x - 1, y - 1)$ and $(x - 1, y + 1)$.

- $(x, y) \notin T$ by assumption.
- $(x-1, y+1)$ is adjacent to $(x, y+1)$, which is in T by (i). By (K1), $(x-1, y+1) \notin T$.
- $(x-1, y-1)$ is adjacent to $(x, y-1) \in T$, so $(x-1, y-1) \notin T$.

(iii) Hence $(x-1, y)$, which is not in T, has at most one neighbour in T, namely possibly $(x-2, y)$. This contradicts (K2).

(M2) for columns. The map $(x, y) \mapsto (y, x)$ is an automorphism of Γ; it maps T to a 2-d kernel whose rows are the columns of T. Applying the previous paragraph to that kernel gives (M2) for the columns of T. □

Lemma 4.3 (adjacent rows)**.** *Let T be a 2-d kernel of Γ and $y \in \mathbb{Z}$. Then*

(a) $T_y \cap T_{y+1} = \emptyset$, *and*

(b) $T_{y-1} \cup T_y \cup T_{y+1} = \mathbb{Z}$.

Proof. Fix $x \in \mathbb{Z}$. By Lemma 4.2 the column T^x is an MIS of $\mathbb{Z}$. By (M1) for T^x, not both y and $y+1$ lie in T^x; that is, not both $x \in T_y$ and $x \in T_{y+1}$. This is (a) at x. By (M2) for T^x, at least one of $y-1, y, y+1$ lies in T^x; that is, $x \in T_{y-1} \cup T_y \cup T_{y+1}$. This is (b) at x. □

4.2 Offsets of consecutive rows

Throughout this subsection T is a 2-d kernel of Γ. By Lemma 4.2 every row is an MIS of $\mathbb{Z}$, and by Lemma 4.3(a), applied at y and at $y-1$, the rows T_{y+1} and T_{y-1} are both disjoint from T_y. So Lemma 3.4 applies with $A = T_y$ and $B = T_{y+1}$, and also with $A = T_y$ and $B = T_{y-1}$.

Definition 4.4 (offsets)**.** Fix a height y and an enumeration (p_i) of T_y with gaps g_i. For each i, let e_i and f_i be the integers with

$$p_i + e_i \in T_{y+1} \cap (p_i, p_{i+1}), \qquad p_i + f_i \in T_{y-1} \cap (p_i, p_{i+1}).$$

By Lemma 3.4 they exist and are unique, and

$$e_i, f_i \in \{1, 2\}, \qquad e_i \le g_i - 1, \qquad f_i \le g_i - 1. \tag{1}$$

We call $e = (e_i)$ and $f = (f_i)$ the *upper* and *lower offsets* of row y with respect to (p_i). By Lemma 3.4, $(p_i + e_i)_i$ enumerates T_{y+1} and $(p_i + f_i)_i$ enumerates T_{y-1}.

If (p_i) is replaced by another enumeration (p_{i+c}) of T_y (Lemma 3.2(c)), the offsets become (e_{i+c}) and (f_{i+c}). In particular the statements "$e_i = 1$ for all i" and "$f_i = 1$ for all i" do not depend on the enumeration. We abbreviate them as "$e \equiv 1$ in row y" and "$f \equiv 1$ in row y".

Lemma 4.5 (gap identity)**.** *With the notation of Definition 4.4, $g_i = e_i + f_i$ for every i.*

Proof. By Lemma 3.3, $g_i \in \{2, 3\}$.

Case $g_i = 2$. By (1), $1 \le e_i \le g_i - 1 = 1$ and likewise for f_i, so $e_i = f_i = 1$ and $e_i + f_i = 2 = g_i$.

Case $g_i = 3$. Let $I = \{p_i + 1, p_i + 2\}$, the set of integers strictly between p_i and p_{i+1}.

- No element of I lies in T_y (Lemma 3.2(a)).
- By the uniqueness in Lemma 3.4, $T_{y+1} \cap I = \{p_i + e_i\}$ and $T_{y-1} \cap I = \{p_i + f_i\}$.

- By Lemma 4.3(b), every element of I lies in $T_{y-1} \cup T_y \cup T_{y+1}$, hence in $T_{y-1} \cup T_{y+1}$.

Therefore $I = (T_{y-1} \cap I) \cup (T_{y+1} \cap I) = \{p_i + f_i, p_i + e_i\}$. Since I has two elements, $\{e_i, f_i\} = \{1, 2\}$, and $e_i + f_i = 3 = g_i$. □

Lemma 4.6 (transport to the next row)**.** *With the notation of Definition 4.4, let $q_i = p_i + e_i$. Then (q_i) is an enumeration of T_{y+1}, and the upper and lower offsets e' and f' of row $y+1$ with respect to (q_i) are*

$$f'_i = f_i \qquad \text{and} \qquad e'_i = e_{i+1} \qquad (i \in \mathbb{Z}).$$

Proof. (q_i) enumerates T_{y+1} by Definition 4.4. The lower offsets of row $y+1$ locate the points of row $(y+1)-1 = y$ inside the gaps of T_{y+1}.

Lower offsets. Using $e_i \le g_i - 1$ and $e_{i+1} \ge 1$,

$$q_i = p_i + e_i \le p_i + g_i - 1 < p_i + g_i = p_{i+1} < p_{i+1} + e_{i+1} = q_{i+1}.$$

So $p_{i+1} \in T_y$ lies in (q_i, q_{i+1}), and by the uniqueness in Lemma 3.4 it is the point of T_y in that interval. Hence, using Lemma 4.5,

$$f'_i = p_{i+1} - q_i = (p_i + g_i) - (p_i + e_i) = g_i - e_i = f_i.$$

Upper offsets. The gaps of T_{y+1} with respect to (q_i) are

$$h_i = q_{i+1} - q_i = (p_{i+1} + e_{i+1}) - (p_i + e_i) = g_i + e_{i+1} - e_i = (e_i + f_i) + e_{i+1} - e_i = f_i + e_{i+1}.$$

Lemma 4.5 applied to row $y+1$ gives $h_i = e'_i + f'_i = e'_i + f_i$. Comparing, $e'_i = e_{i+1}$. □

Corollary 4.7 (iterated transport)**.** *With the notation of Definition 4.4, let $t \ge 0$. There is an enumeration of T_{y+t} with respect to which the upper and lower offsets of row $y+t$ are $(e_{i+t})_{i\in\mathbb{Z}}$ and $(f_i)_{i\in\mathbb{Z}}$, and the gaps of T_{y+t} are $e_{i+t} + f_i$. Consequently:*

(a) $e \equiv 1$ in row y if and only if $e \equiv 1$ in row $y+t$;

(b) $f \equiv 1$ in row y if and only if $f \equiv 1$ in row $y+t$.

Proof. Induction on t. For $t = 0$ take (p_i) itself. If the claim holds for t, apply Lemma 4.6 to row $y+t$ with the enumeration given by the induction hypothesis: the new offsets are $(e_{(i+1)+t})_i = (e_{i+(t+1)})_i$ and $(f_i)_i$. The gaps follow from Lemma 4.5. For (a), $(e_{i+t})_i$ is the sequence $(e_i)_i$ re-indexed, so it is constantly 1 if and only if (e_i) is. For (b), the lower offsets are literally the same sequence. □

Let $y < y'$ be any two heights. Applying Corollary 4.7 with base row y and $t = y' - y$ shows that $e \equiv 1$ holds in row y exactly when it holds in row y', and the same for $f \equiv 1$. Hence:

$$\text{each of the properties “}e \equiv 1\text{” and “}f \equiv 1\text{” holds either in every row or in no row.} \tag{2}$$

Lemma 4.8 (dichotomy)**.** *In every row, $e \equiv 1$ or $f \equiv 1$.*

Proof. Suppose not. Then there are a row y, an enumeration of T_y with offsets (e, f), and indices i, j with $e_j = 2$ and $f_i = 2$.

Case $j \ge i$. Apply Corollary 4.7 with $t = j - i \ge 0$. Row $y+t$ has the gap $e_{i+t} + f_i = e_j + f_i = 2 + 2 = 4$, which contradicts Lemma 3.3.

Case $j < i$. Put $t = i - j > 0$, choose an enumeration of T_{y-t}, and let (e^*, f^*) be its offsets. By Corollary 4.7 with base row $y - t$, there is an enumeration (r_k) of T_y whose offsets are $(e^*_{k+t})_k$ and $(f^*_k)_k$. By Lemma 3.2(c), the fixed enumeration of T_y is $(r_{k+c})_k$ for some $c \in \mathbb{Z}$, so its offsets are $e_k = e^*_{k+c+t}$ and $f_k = f^*_{k+c}$ for all k. Then

$$e^*_{i+c} = e^*_{(j+t)+c} = e_j = 2 \qquad \text{and} \qquad f^*_{i+c} = f_i = 2,$$

so by Lemma 4.5 row $y - t$ has the gap $e^*_{i+c} + f^*_{i+c} = 4$, which again contradicts Lemma 3.3. □

4.3 Classification

Theorem 4.9 (2-d kernels of the grid)**.** *A set $T \subseteq \mathbb{Z}^2$ is a 2-d kernel of Γ if and only if $T = D_s(K)$ for some $s \in \{+1, -1\}$ and some MIS K of $\mathbb{Z}$. Moreover, K is determined by T, namely $K = T_0$; equivalently, for fixed s the map $K \mapsto D_s(K)$ is injective.*

Proof. ($\Leftarrow$) Let $T = D_s(K)$. Its rows are $T_y = K + sy$ (Definition 4.1). Translation $z \mapsto z + sy$ is an automorphism of the path on $\mathbb{Z}$, so it maps the MIS K to an MIS. Its columns are

$$T^x = \{y : x - sy \in K\} = \{y : sy \in x - K\} = \{sz : z \in x - K\} = s\,(x - K),$$

where we used $s^{-1} = s$. The maps $z \mapsto x - z$ and $z \mapsto sz$ are automorphisms of the path on $\mathbb{Z}$, so T^x is an MIS. By Lemma 4.2, T is a 2-d kernel.

($\Rightarrow$) Let T be a 2-d kernel. By (2), either $e \equiv 1$ in every row, or $e \equiv 1$ in no row.

(i) $e \equiv 1$ *in every row.* For each y, with any enumeration (p_i) of T_y, $T_{y+1} = \{p_i + e_i\} = \{p_i + 1\} = T_y + 1$ (Definition 4.4). By induction, $T_y = T_0 + y$ for $y \geq 0$, and also for $y < 0$, because $T_y = T_{y+1} - 1$. Hence $(x, y) \in T \iff x \in T_0 + y \iff x - y \in T_0$, that is, $T = D_{+1}(T_0)$.

(ii) $e \equiv 1$ *in no row.* By Lemma 4.8, $f \equiv 1$ in every row. So $T_{y-1} = \{p_i + f_i\} = T_y + 1$ for every y, that is, $T_y = T_{y-1} - 1$. By induction in both directions, $T_y = T_0 - y$ for all y, so $(x, y) \in T \iff x + y \in T_0$, that is, $T = D_{-1}(T_0)$.

In both cases $K = T_0$ is an MIS of $\mathbb{Z}$ by Lemma 4.2. Finally, the row at height 0 of $D_s(K)$ is $K + 0 = K$, so K is recovered from $D_s(K)$. □

Lemma 4.10 (the two families meet only in checkerboards)**.** *Let K, K' be MIS of $\mathbb{Z}$. Then $D_{+1}(K) = D_{-1}(K')$ if and only if $K = K'$ and $K \in \{2\mathbb{Z},\ 2\mathbb{Z} + 1\}$.*

Proof. ($\Rightarrow$) The rows at height 0 are K and K', so $K = K'$. The rows at height 1 are $K + 1$ and $K' - 1 = K - 1$, so $K + 1 = K - 1$, and adding 1 gives $K + 2 = K$. Thus K is a 2-periodic MIS, and $K \in \{2\mathbb{Z}, 2\mathbb{Z} + 1\}$ by Lemma 3.7.

($\Leftarrow$) If $K + 2 = K$, then $K + 2y = K$ for every $y \in \mathbb{Z}$, hence $K + y = K - y$ (add $-y$ to both sides of $K + 2y = K$). So $D_{+1}(K)$ and $D_{-1}(K)$ have the same rows, and they are equal. □

5 Quotients of the grid

Definition 5.1 (admissible lattice, quotient graph)**.** Let $U = \{(1, 0), (-1, 0), (0, 1), (0, -1)\}$ and

$$F = U \cup \{(\pm 2, 0),\ (0, \pm 2),\ \pm(1, 1),\ \pm(1, -1)\}.$$

A subgroup $L \leq \mathbb{Z}^2$ is an *admissible lattice* if it has finite index in $\mathbb{Z}^2$ and $L \cap F = \emptyset$. The *quotient graph* G_L has vertex set $\mathbb{Z}^2/L$, and two cosets $[v], [w]$ are adjacent if $w - v \in U + L$. We write $\pi : \mathbb{Z}^2 \to \mathbb{Z}^2/L$ for the quotient map, $\pi(v) = [v] = v + L$.

The adjacency is well defined: if $v' = v + \ell$ and $w' = w + \ell'$ with $\ell, \ell' \in L$, then $w' - v' = (w - v) + (\ell' - \ell)$, and since $U + L$ is invariant under adding elements of L, $w' - v' \in U + L \iff w - v \in U + L$. Note also that $F \setminus U$ is exactly the set of nonzero differences $u - u'$ of elements $u, u' \in U$: the differences are $(\pm 2, 0)$, $(0, \pm 2)$ and $(\pm 1, \pm 1)$.

Lemma 5.2 (covering)**.** *Let L be admissible. Then G_L is a simple 4-regular graph, and for every $v \in \mathbb{Z}^2$ the map π restricts to a bijection from $N_\Gamma(v) = v + U$ onto $N_{G_L}(\pi(v))$.*

Proof. Symmetry. $U = -U$, so $U + L = -(U + L)$, and $w - v \in U + L$ implies $v - w \in U + L$.

No loops. $[v] \sim [v]$ would mean $0 \in U + L$, that is, $-u \in L$ for some $u \in U$; since $-u \in U \subseteq F$, this contradicts $L \cap F = \emptyset$.

Neighbourhoods. $[w] \sim [v]$ if and only if $w = v + u + \ell$ for some $u \in U$ and $\ell \in L$, if and only if $[w] = [v + u]$ for some $u \in U$. So $N_{G_L}([v]) = \{[v + u] : u \in U\} = \pi(v + U)$, and π maps $v + U$ onto $N_{G_L}(\pi(v))$.

Injectivity. If $[v + u] = [v + u']$ for distinct $u, u' \in U$, then $u - u' \in L$, and $u - u' \in F \setminus U$ by the remark above, contradicting $L \cap F = \emptyset$. So π is injective on $v + U$, and $|N_{G_L}([v])| = 4$. □

Lemma 5.3 (lifting)**.** *Let L be admissible and $S \subseteq \mathbb{Z}^2/L$. Then S is a 2-d kernel of G_L if and only if $\pi^{-1}(S)$ is a 2-d kernel of Γ. Consequently $S \mapsto \pi^{-1}(S)$ is a bijection from the set of 2-d kernels of G_L onto the set of L-invariant 2-d kernels of Γ, with inverse $T \mapsto \pi(T)$.*

Proof. Write $\widetilde{S} = \pi^{-1}(S)$. For every $v \in \mathbb{Z}^2$:

- $v \in \widetilde{S}$ if and only if $\pi(v) \in S$, by definition of the preimage;
- $|N_\Gamma(v) \cap \widetilde{S}| = |N_{G_L}(\pi(v)) \cap S|$, because by Lemma 5.2 π is a bijection from $N_\Gamma(v)$ onto $N_{G_L}(\pi(v))$, and a neighbour w of v lies in $\widetilde{S}$ exactly when $\pi(w) \in S$.

A set is a 2-d kernel exactly when every vertex u satisfies: if u is in the set then it has 0 neighbours in the set, and if u is not in the set then it has at least 2. By the two bullets, this condition holds at v for $\widetilde{S}$ in Γ if and only if it holds at $\pi(v)$ for S in G_L. Every vertex of G_L has the form $\pi(v)$, so $\widetilde{S}$ is a 2-d kernel of Γ if and only if S is one of G_L.

For the bijection: $\pi^{-1}(S)$ is L-invariant, and $\pi(\pi^{-1}(S)) = S$ because π is surjective. Conversely, if T is L-invariant then $\pi^{-1}(\pi(T)) = T$: the inclusion $\supseteq$ is clear, and if $\pi(v) \in \pi(T)$ then $v = t + \ell$ for some $t \in T$ and $\ell \in L$, so $v \in T$. Applying the first part to $S = \pi(T)$ shows that $\pi(T)$ is a 2-d kernel of G_L when T is an L-invariant 2-d kernel of Γ. □

Definition 5.4 (diagonal periods)**.** Let L be an admissible lattice and $s \in \{+1, -1\}$. The image $\varphi_s(L)$ is a subgroup of $\mathbb{Z}$, so $\varphi_s(L) = d_s\mathbb{Z}$ for a unique $d_s \geq 0$ (Fact 2.1). We write $d_+ = d_{+1}$ and $d_- = d_{-1}$.

Lemma 5.5. *For an admissible lattice, $d_+ \geq 1$ and $d_- \geq 1$.*

Proof. We have $\varphi_s(x, y) = 0$ if and only if $x = sy$, so $\ker \varphi_s = \mathbb{Z} \cdot (s, 1)$. Suppose $d_s = 0$, i.e. $L \subseteq \mathbb{Z} \cdot (s, 1)$. The vertices $(k, 0)$, $k \in \mathbb{Z}$, lie in pairwise different cosets of $\mathbb{Z} \cdot (s, 1)$, because $(k - k', 0) = y(s, 1)$ forces $y = 0$ and $k = k'$. Hence they lie in pairwise different cosets of the smaller group L, and L has infinite index, a contradiction. □

Theorem 5.6 (main theorem)**.** *Let L be an admissible lattice and $N = |\mathbb{Z}^2/L|$.*

(a) The 2-d kernels of G_L are exactly the sets $\pi(D_s(K))$ with $s \in \{+1, -1\}$ and K a d_s-periodic MIS of $\mathbb{Z}$.

(b) The number of 2-d kernels of G_L is

$$\sigma(G_L) = P(d_+) + P(d_-) - 2\,[d_+ \text{ and } d_- \text{ are both even}].$$

(c) G_L has a 2-d kernel if and only if $d_+ \geq 2$ or $d_- \geq 2$.

(d) The sizes of the 2-d kernels of G_L are exactly the numbers Nk/d_s with $s \in \{+1, -1\}$, $d_s \geq 2$ and $\lceil d_s/3 \rceil \leq k \leq \lfloor d_s/2 \rfloor$.

Proof. (a) By Lemma 5.3, the 2-d kernels of G_L are the sets $\pi(T)$ with T an L-invariant 2-d kernel of Γ, and by Theorem 4.9 these T are the sets $D_s(K)$ that are L-invariant. For $\ell \in L$,

$$D_s(K)+\ell = \{v+\ell : \varphi_s(v) \in K\} = \{w : \varphi_s(w-\ell) \in K\} = \{w : \varphi_s(w) \in K+\varphi_s(\ell)\} = D_s(K+\varphi_s(\ell)),$$

using that φ_s is additive. By the injectivity in Theorem 4.9, $D_s(K) + \ell = D_s(K)$ if and only if $K + \varphi_s(\ell) = K$. So $D_s(K)$ is L-invariant if and only if $K + t = K$ for every $t \in \varphi_s(L) = d_s\mathbb{Z}$. This holds if and only if $K + d_s = K$: one direction takes $t = d_s$, and conversely $K + d_s = K$ implies $K + td_s = K$ for all $t \in \mathbb{Z}$ (see the remark after Definition 3.5).

(b) By (a), Lemma 5.3 (which says distinct L-invariant sets have distinct images) and Theorem 4.9 (injectivity of $K \mapsto D_s(K)$), the number of 2-d kernels of G_L is the number of distinct sets in the union of the two families

$$\mathcal{F}_+ = \{D_{+1}(K) : K \text{ a } d_+\text{-periodic MIS}\}, \qquad \mathcal{F}_- = \{D_{-1}(K) : K \text{ a } d_-\text{-periodic MIS}\},$$

which have $M(d_+)$ and $M(d_-)$ members respectively. By Lemma 4.10, a set lies in both families exactly when it equals $D_{+1}(K) = D_{-1}(K)$ with $K \in \{2\mathbb{Z}, 2\mathbb{Z}+1\}$ and K both d_+-periodic and d_--periodic. The set $2\mathbb{Z}$ is d-periodic if and only if d is even, and the same holds for $2\mathbb{Z}+1$. So $|\mathcal{F}_+ \cap \mathcal{F}_-| = 2$ if d_+ and d_- are both even and 0 otherwise. By inclusion and exclusion and Lemma 3.7, $\sigma(G_L) = M(d_+) + M(d_-) - |\mathcal{F}_+ \cap \mathcal{F}_-|$, which is the stated formula.

(c) Both $d_\pm \geq 1$ by Lemma 5.5, $P(1) = 0$, and $P(d) \geq 2$ for $d \geq 2$ (Lemma 3.9).

- If $d_+ = d_- = 1$, the count is $0 + 0 - 0 = 0$.
- If exactly one of them is at least 2, the other equals 1, which is odd, so the correction term is 0 and the count is at least 2.
- If both are at least 2, the count is at least $2 + 2 - 2 = 2$.

(d) The map $\bar{\varphi}_s : \mathbb{Z}^2/L \to \mathbb{Z}_{d_s}$, $[v] \mapsto q_{d_s}(\varphi_s(v))$, is well defined, because $v - v' \in L$ gives $\varphi_s(v) - \varphi_s(v') = \varphi_s(v - v') \in d_s\mathbb{Z}$. It is a homomorphism because φ_s is additive, and it is surjective because $\varphi_s(t, 0) = t$ for every $t \in \mathbb{Z}$. By Fact 2.4 each fibre of $\bar{\varphi}_s$ has N/d_s elements. Let K be d_s-periodic. Then

$$[v] \in \pi(D_s(K)) \iff \exists\, \ell \in L : \varphi_s(v + \ell) \in K \iff \varphi_s(v) \in K \iff \bar{\varphi}_s([v]) \in \overline{K},$$

where the middle step holds because $\varphi_s(v+\ell) - \varphi_s(v) \in d_s\mathbb{Z}$ and K is d_s-periodic. Hence $\pi(D_s(K)) = \bar{\varphi}_s^{-1}(\overline{K})$, which has $|\overline{K}| \cdot N/d_s$ elements. The possible values of $|\overline{K}|$ are given by Lemma 3.10 when $d_s \geq 2$; when $d_s = 1$ there is no K (Lemma 3.7). □

Remark 5.7. The hypothesis $L \cap F = \emptyset$ is used only in Lemma 5.2. Without it the quotient has loops or parallel edges, or two grid neighbours of a vertex collapse to one vertex, and the neighbour count in Lemma 5.3 breaks down.

6 Tori, tensor products and circulants

6.1 Tori

Corollary 6.1 (tori)**.** *Let $m, n \geq 3$ and $d = \gcd(m, n)$. Then:*

(a) $C_m \,\square\, C_n$ has a 2-d kernel if and only if $d \geq 2$;

(b) every 2-d kernel is the set $\{(x,y) \in \mathbb{Z}_m \times \mathbb{Z}_n : (x - sy) \bmod d \in \overline{K}\}$ *for some* $s \in \{+1,-1\}$ *and some maximal independent set* $\overline{K}$ *of* C_d *(for* $d = 2$*:* $\overline{K} \in \{\{0\},\{1\}\}$*), and every such set is a 2-d kernel;*

(c) $\sigma(C_m \,\square\, C_n) = 2P(d) - 2\,[\,d$ *even*$\,]$*;*

(d) the sizes of the 2-d kernels are exactly mnk/d *for* $\lceil d/3 \rceil \le k \le \lfloor d/2 \rfloor$*; in particular the minimum size of a 2-d kernel is* $mn\lceil d/3 \rceil / d$*.*

Proof. Let $L = m\mathbb{Z} \times n\mathbb{Z} = \{(am, bn) : a, b \in \mathbb{Z}\}$, a subgroup of index mn.

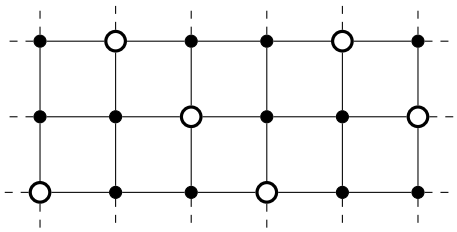
$G_L \cong C_m \,\square\, C_n$. The map $\mathbb{Z}^2/L \to \mathbb{Z}_m \times \mathbb{Z}_n$, $[(x,y)] \mapsto (x \bmod m, y \bmod n)$, is well defined and bijective. Two cosets are adjacent in G_L when their difference is congruent modulo (m,n) to one of $(\pm 1, 0)$, $(0, \pm 1)$, which is exactly adjacency in $C_m \,\square\, C_n$.

Admissibility. We check that no element of F lies in L. An element (am, bn) of L has first coordinate divisible by $m \ge 3$ and second coordinate divisible by $n \ge 3$. Every element of F, namely $(\pm 1, 0)$, $(0, \pm 1)$, $(\pm 2, 0)$, $(0, \pm 2)$, $\pm(1,1)$ and $\pm(1,-1)$, has at least one coordinate whose absolute value is 1 or 2. An integer of absolute value 1 or 2 is not divisible by any integer that is at least 3, so that coordinate is not divisible by m (if it is the first) or by n (if it is the second). So $L \cap F = \emptyset$.

Periods. $\varphi_s(am, bn) = am - sbn$. As a, b range over $\mathbb{Z}$ this ranges over $m\mathbb{Z} + n\mathbb{Z} = d\mathbb{Z}$ (Fact 2.2; the sign s is irrelevant because b ranges over $\mathbb{Z}$). So $d_+ = d_- = d$.

Conclusions. (a) is Theorem 5.6(c). (c) is Theorem 5.6(b), since d_+ and d_- are both even exactly when d is even. (d) is Theorem 5.6(d) with $N = mn$. For (b): by Theorem 5.6(a) the kernels are $\pi(D_s(K))$ with K a d-periodic MIS, and by the computation in the proof of Theorem 5.6(d), $\pi(D_s(K)) = \{[v] : \varphi_s(v) \bmod d \in \overline{K}\}$, which in the coordinates of $\mathbb{Z}_m \times \mathbb{Z}_n$ reads $\{(x,y) : (x - sy) \bmod d \in \overline{K}\}$ (well defined since $d \mid m$ and $d \mid n$). The sets $\overline{K}$ are the maximal independent sets of C_d for $d \ge 3$ (Lemma 3.6) and $\{0\}, \{1\}$ for $d = 2$ (Lemma 3.7). □

Example 6.2. In $C_6 \,\square\, C_3$ ($m = 6$, $n = 3$, $d = 3$) take $s = -1$ and $\overline{K} = \{0\}$, the maximal independent set of C_3 consisting of one vertex. The kernel is $S = \{(x,y) : x + y \equiv 0 \pmod 3\}$, shown in Figure 1. We verify it directly. Two adjacent vertices differ by ± 1 in one coordinate, so their values of $x + y$ differ by ± 1 modulo 3 and cannot both be 0; so S is independent. If $x + y \equiv 1$, the neighbours $(x - 1, y)$ and $(x, y - 1)$ have $x + y - 1 \equiv 0$; if $x + y \equiv 2$, the neighbours $(x + 1, y)$ and $(x, y + 1)$ have $x + y + 1 \equiv 0$. So every vertex outside S has exactly two neighbours in S. In total $\sigma(C_6 \,\square\, C_3) = 2P(3) - 0 = 6$: for each $s = \pm 1$, the three choices $\overline{K} = \{0\}, \{1\}, \{2\}$.



$x = 0, \ldots, 5$ (horizontal), $y = 0, 1, 2$ (vertical)

Figure 1: The 2-d kernel $\{x + y \equiv 0 \pmod 3\}$ of $C_6 \,\square\, C_3$ (white vertices). Dashed edges wrap around. According to [1, Theorem 2.5] this graph has no 2-d kernel.

6.2 Tensor products of cycles

By Definition 1.3, the tensor product $C_m \times C_n$ has vertex set $\mathbb{Z}_m \times \mathbb{Z}_n$, and (i, j) is adjacent to $(i + \alpha, j + \beta)$ for all $\alpha, \beta \in \{+1, -1\}$.

Corollary 6.3 (tensor products)**.** *Let $m, n \geq 3$. Then*

$$\sigma(C_m \times C_n) = \begin{cases} P(m) + P(n) & \text{if } m \text{ and } n \text{ are odd,} \\ P(2m) + P(n) - 2 & \text{if } m \text{ is odd and } n \text{ is even,} \\ P(m) + P(2n) - 2 & \text{if } m \text{ is even and } n \text{ is odd,} \\ \big(P(m) + P(n) - 2\big)^2 & \text{if } m \text{ and } n \text{ are even.} \end{cases}$$

In particular $C_m \times C_n$ always has a 2-d kernel, and $\sigma(C_m \times C_3) = P(m) + 3$ for every $m \geq 3$.

Proof. We set up notation in four steps and then treat the cases.

Step 1: the even sublattice. Let $E = \{(i, j) \in \mathbb{Z}^2 : i \equiv j \pmod 2\}$, a subgroup of index 2 in $\mathbb{Z}^2$ with cosets E and $(1, 0) + E$. Let $W = \{(\alpha, \beta) : \alpha, \beta \in \{+1, -1\}\} \subseteq E$. For a subgroup $M \leq E$ of finite index, let H_M be the graph on E/M in which $[a] \sim [b]$ if $b - a \in W + M$ (well defined by the argument after Definition 5.1).

Step 2: rotating the grid. The map $\psi : \mathbb{Z}^2 \to E$, $\psi(u, v) = (u+v, u-v)$, takes values in E because $(u + v) - (u - v) = 2v$ is even. It is additive, and it is bijective with inverse $(i, j) \mapsto \left(\frac{i+j}{2}, \frac{i-j}{2}\right)$, which has integer values on E. It maps

$$(1, 0) \mapsto (1, 1), \quad (-1, 0) \mapsto (-1, -1), \quad (0, 1) \mapsto (1, -1), \quad (0, -1) \mapsto (-1, 1),$$

so $\psi(U) = W$. For a finite-index subgroup $M \leq E$ put $L = \psi^{-1}(M)$. Then ψ induces a bijection $\mathbb{Z}^2/L \to E/M$, $[v] \mapsto [\psi(v)]$, and $w - v \in U + L$ if and only if $\psi(w) - \psi(v) = \psi(w - v) \in W + M$. Hence $G_L \cong H_M$. Moreover, for $(i, j) \in E$ and $(u, v) = \psi^{-1}(i, j)$,

$$\varphi_{+1}(u, v) = u - v = j, \qquad \varphi_{-1}(u, v) = u + v = i.$$

So d_+ is the generator of the group of j-coordinates of elements of M, and d_- the generator of the group of i-coordinates.

Step 3: admissibility. Let $\Lambda = m\mathbb{Z} \times n\mathbb{Z}$, and suppose $M \subseteq \Lambda$. The images under ψ of the elements of F are

$$\psi(\pm(1,0)) = \pm(1,1),\ \psi(\pm(0,1)) = \pm(1,-1),\ \psi(\pm(2,0)) = \pm(2,2),\ \psi(\pm(0,2)) = \pm(2,-2),\ \psi(\pm(1,1)) = \pm(2,0),\ \psi$$

Each of these has a coordinate equal to ± 1 or ± 2 in a position where elements of Λ have coordinates divisible by $m \geq 3$ or $n \geq 3$: the first coordinate in the first four cases and in $\pm(2, 0)$, and the second coordinate in $\pm(0, 2)$. So none of them lies in $\Lambda \supseteq M$, and $L \cap F = \emptyset$. Since Λ has finite index in $\mathbb{Z}^2$, M has finite index in E in both cases below, so L is admissible.

Step 4: the tensor product as a quotient. By definition $C_m \times C_n$ is the graph on $\mathbb{Z}^2/\Lambda$ in which $[a] \sim [b]$ if $b - a \in W + \Lambda$.

Case 1: m or n is odd. Then $\Lambda \not\subseteq E$: if m is odd then $(m, 0) \in \Lambda \setminus E$, and if n is odd then $(0, n) \in \Lambda \setminus E$. Take $\lambda \in \Lambda \setminus E$; then $\lambda + E = (1, 0) + E$, so $\mathbb{Z}^2 = E \cup (\lambda + E) \subseteq E + \Lambda$. Let $M = E \cap \Lambda$. The map $E/M \to \mathbb{Z}^2/\Lambda$, $a + M \mapsto a + \Lambda$, is well defined (as $M \subseteq \Lambda$), injective (if $a - a' \in \Lambda$ with $a, a' \in E$, then $a - a' \in E \cap \Lambda = M$), and surjective (every $z \in \mathbb{Z}^2$ is $a + \lambda'$ with $a \in E$ and $\lambda' \in \Lambda$). It preserves adjacency in both directions: for $a, b \in E$ we have $b - a \in E$ and $W \subseteq E$, so $b - a \in W + \Lambda$ if and only if $b - a \in W + (E \cap \Lambda)$. Hence $C_m \times C_n \cong H_M \cong G_L$ with $L = \psi^{-1}(M)$, and $\sigma(C_m \times C_n) = \sigma(G_L)$. Now $M = \{(am, bn) : am \equiv bn \pmod 2\}$.

- *m and n odd.* Then $am \equiv a$ and $bn \equiv b \pmod 2$, so M consists of the (am, bn) with $a \equiv b \pmod 2$. For every a the choice $b = a$ is allowed, so the i-coordinates of M are all of $m\mathbb{Z}$; symmetrically the j-coordinates are all of $n\mathbb{Z}$. Hence $d_- = m$ and $d_+ = n$, both odd, and Theorem 5.6(b) gives $P(m) + P(n)$.

- *m odd and n even.* Then bn is even, so the condition is that am is even, that is, a is even; b is arbitrary. The i-coordinates form $2m\mathbb{Z}$ and the j-coordinates form $n\mathbb{Z}$. Hence $d_- = 2m$ and $d_+ = n$, both even, and Theorem 5.6(b) gives $P(2m) + P(n) - 2$.

- *m even and n odd.* By the symmetric argument, $d_- = m$ and $d_+ = 2n$, both even, giving $P(m) + P(2n) - 2$.

Case 2: m and n even. Then $\Lambda \subseteq E$. Every coset of Λ lies inside one coset of E, so $\mathbb{Z}^2/\Lambda$ is the disjoint union of $V_0 = E/\Lambda$ and $V_1 = ((1,0) + E)/\Lambda$. Since $W \subseteq E$, every edge joins two vertices in the same part. Translation $a \mapsto a + (1,0)$ maps E bijectively onto $(1,0) + E$ and preserves differences, so it induces an isomorphism between the subgraphs induced on V_0 and V_1. The subgraph on V_0 is $H_\Lambda \cong G_L$ with $L = \psi^{-1}(\Lambda)$; here $M = \Lambda$ has i-coordinates $m\mathbb{Z}$ and j-coordinates $n\mathbb{Z}$, so $d_- = m$ and $d_+ = n$, both even, and the part has $P(m) + P(n) - 2$ kernels. In a disjoint union of two graphs with no edges between them, every neighbourhood lies inside one part, so a set is a 2-d kernel if and only if its intersection with each part is a 2-d kernel of that part. Hence $\sigma(C_m \times C_n) = (P(m) + P(n) - 2)^2$.

Existence. In every case at least one of $d_\pm$ is at least $m \geq 3$ or $n \geq 3$, so a 2-d kernel exists by Theorem 5.6(c) (in Case 2, in each part).

The case $n = 3$. If m is odd, $P(m) + P(3) = P(m) + 3$. If m is even, the third line applies with $n = 3$: $P(m) + P(6) - 2 = P(m) + 5 - 2 = P(m) + 3$. □

Remark 6.4. For $n = 3$, Corollary 6.3 is [2, Theorem 9(i)] (stated there as $\sigma(C_n \times C_3) = 3 + P(n)$), and the existence statement contains [2, Theorem 6(ii)]. When $m = n$ is odd, $M = \{(an, bn) : a \equiv b \pmod 2\}$ and $\psi^{-1}(an, bn) = \left(\frac{a+b}{2}n, \frac{a-b}{2}n\right)$; as $a \equiv b$ ranges over all pairs of the same parity, $(\frac{a+b}{2}, \frac{a-b}{2})$ ranges over all of $\mathbb{Z}^2$. So $L = n\mathbb{Z} \times n\mathbb{Z}$ and $C_n \times C_n \cong C_n \,\square\, C_n$ for odd n; accordingly Corollaries 6.1 and 6.3 both give $2P(n)$.

6.3 Circulants

For $N \geq 1$ and $a, b \in \mathbb{Z}$, the circulant $C_N(a, b)$ has vertex set $\mathbb{Z}_N$, and k is adjacent to $k \pm a$ and $k \pm b$.

Corollary 6.5 (circulants)**.** *Let $N \geq 1$ and $a, b \in \mathbb{Z}$ with $\gcd(a, b, N) = 1$, and suppose that none of $a, b, 2a, 2b, a+b, a-b$ is divisible by N. Then $C_N(a, b)$ is a connected simple 4-regular graph, and*

$$\sigma(C_N(a,b)) = P(g_+) + P(g_-) - 2\,[\,g_+, g_- \text{ both even}\,], \qquad g_\pm = \gcd(N, a \pm b).$$

In particular $C_N(a, b)$ has a 2-d kernel if and only if $\gcd(N, a + b) > 1$ or $\gcd(N, a - b) > 1$.

Proof. The quotient. Let $\chi : \mathbb{Z}^2 \to \mathbb{Z}_N$, $\chi(x, y) = q_N(xa + yb)$, and $L = \ker \chi$. The image of χ is the subgroup of $\mathbb{Z}_N$ generated by $q_N(a)$ and $q_N(b)$; by Fact 2.2 there are integers t_1, t_2, t_3 with $t_1 a + t_2 b + t_3 N = \gcd(a, b, N) = 1$, so $q_N(1) = \chi(t_1, t_2)$ lies in the image, and χ is surjective. Hence χ induces a group isomorphism $\bar{\chi} : \mathbb{Z}^2/L \to \mathbb{Z}_N$, and L has index N. Under $\bar{\chi}$, $[w] \sim [v]$ in G_L (that is, $w - v \in U + L$) if and only if $\chi(w) - \chi(v) \in \chi(U) = \{\pm a, \pm b\}$ (modulo N), which is adjacency in $C_N(a, b)$. So $G_L \cong C_N(a, b)$.

Admissibility. χ maps the elements of F to $\pm a, \pm b, \pm 2a, \pm 2b, \pm(a+b), \pm(a-b)$. By hypothesis none of these is 0 in $\mathbb{Z}_N$, so $L \cap F = \emptyset$. By Lemma 5.2, $C_N(a,b)$ is simple and 4-regular; it is connected because $\mathbb{Z}^2 = \Gamma$ is connected and π maps edges to edges.

Periods. Fix $s \in \{+1, -1\}$ and put $g = \gcd(N, a + sb)$. Every $(x, y) \in \mathbb{Z}^2$ can be written as $x = t + sy$ with $t = \varphi_s(x, y)$. Then, using $s^2 = 1$,

$$xa + yb = (t + sy)a + yb = ta + y(sa + b) = ta + sy(a + sb). \tag{3}$$

- $\varphi_s(L) \subseteq g\mathbb{Z}$. Let $(x, y) \in L$, so $N \mid xa+yb$. Since $g \mid N$ and $g \mid a+sb$, (3) gives $g \mid ta$. Suppose a prime r divides both g and a. Then $r \mid a + sb$ gives $r \mid sb$, hence $r \mid b$, and $r \mid g \mid N$; so $r \mid \gcd(a, b, N) = 1$, which is impossible. Thus $\gcd(g, a) = 1$, and $g \mid ta$ gives $g \mid t = \varphi_s(x, y)$.
- $g \in \varphi_s(L)$. By Fact 2.3, the congruence $s(a + sb)\, y \equiv -ga \pmod N$ has a solution y, because $\gcd(s(a + sb), N) = g$ divides $-ga$. Put $x = g + sy$. By (3) with $t = g$, $xa + yb = ga + sy(a + sb) \equiv ga - ga = 0 \pmod N$, so $(x, y) \in L$, and $\varphi_s(x, y) = g$.

Hence $\varphi_s(L) = g\mathbb{Z}$ and $d_s = \gcd(N, a + sb)$, so $d_+ = g_+$ and $d_- = g_-$. Theorem 5.6(b) and (c) give the claims. □

Remark 6.6. If $\gcd(m, n) = 1$, then $C_m \,\square\, C_n \cong C_{mn}(n, m)$ via the Chinese remainder isomorphism $\mathbb{Z}_m \times \mathbb{Z}_n \to \mathbb{Z}_{mn}$, which sends $(1, 0)$ and $(0, 1)$ to elements of orders m and n. Any common prime factor of mn and $m \pm n$ would divide both m and n, so $\gcd(mn, m \pm n) = 1$, and Corollary 6.5 gives no 2-d kernel, in agreement with Corollary 6.1.

7 Three-dimensional tori with a triangle factor

In this section we work with colours in $\mathbb{Z}_3 = \{0, 1, 2\}$, where $2 = -1$. Its nonzero elements are 1 and -1, and for $u, w \in \{1, -1\}$ we have $u = -w$ if and only if $u \neq w$.

Definition 7.1 (fall colouring [5])**.** A *fall* 3*-colouring* of a graph G is a map $f : V(G) \to \mathbb{Z}_3$ such that

(F1) f is a proper colouring: $f(u) \neq f(v)$ for every edge uv, and

(F2) every vertex v has, for each colour $k \neq f(v)$, a neighbour w with $f(w) = k$.

7.1 Reduction to fall colourings

The cycle C_3 is the complete graph K_3: any two distinct elements of $\mathbb{Z}_3$ differ by 1 or -1. So for a graph G, the graph $C_3 \,\square\, G$ has vertex set $\mathbb{Z}_3 \times V(G)$ and

$$N_{C_3 \square G}(x, p) = \{(x', p) : x' \in \mathbb{Z}_3,\ x' \neq x\} \cup \{(x, q) : q \in N_G(p)\}, \tag{4}$$

a set of exactly $2 + |N_G(p)|$ vertices (the two parts are disjoint, since vertices in the first part have second coordinate p and those in the second part have second coordinate in $N_G(p) \not\ni p$). For $p \in V(G)$ the *fibre* $\mathbb{Z}_3 \times \{p\}$ induces a triangle.

Lemma 7.2 (triangle reduction)**.** *Let G be a graph. For a map $f : V(G) \to \mathbb{Z}_3$ put $S_f = \{(f(p), p) : p \in V(G)\}$.*

(a) If f is a fall 3-colouring of G, then S_f is a 2-d kernel of $C_3 \,\square\, G$.

(b) *If every vertex of G has degree at most 5, then every 2-d kernel of $C_3 \,\square\, G$ equals S_f for exactly one map f, and that f is a fall 3-colouring of G.*

Consequently, if every vertex of G has degree at most 5, then $\sigma(C_3 \,\square\, G)$ equals the number of fall 3-colourings of G.

Proof. We first compute, for an arbitrary map f, when S_f is a 2-d kernel. The set S_f meets every fibre in exactly one vertex.

Independence of S_f. Two vertices of S_f lie in different fibres, say $(f(p), p)$ and $(f(q), q)$ with $p \neq q$. By (4) they are adjacent if and only if $pq \in E(G)$ and $f(p) = f(q)$. So S_f is independent if and only if f satisfies (F1).

Domination by S_f. Let $(x, p) \notin S_f$, that is, $x \neq f(p)$. By (4), its neighbours in S_f are

- the vertex $(f(p), p)$, which is a neighbour because $f(p) \neq x$, and
- the vertices (x, q) with $q \in N_G(p)$ and $f(q) = x$.

So $|N(x,p) \cap S_f| = 1 + |\{q \in N_G(p) : f(q) = x\}|$, and this is at least 2 if and only if p has a neighbour of colour x. Requiring this for every p and every $x \neq f(p)$ is exactly (F2).

Hence S_f is a 2-d kernel if and only if f is a fall 3-colouring. This proves (a).

(b) Let S be a 2-d kernel of $C_3 \,\square\, G$, and let $p \in V(G)$.

- *At most one vertex of S in the fibre of p.* The fibre is a triangle, and S is independent.
- *At least one.* Suppose the fibre of p contains no vertex of S. For $x \in \mathbb{Z}_3$, the vertex (x, p) is not in S, and its neighbours (x', p) in the fibre are not in S; so by (4) and (K2) the set $Q_x = \{q \in N_G(p) : (x, q) \in S\}$ has at least 2 elements. The sets Q_0, Q_1, Q_2 are pairwise disjoint: if $q \in Q_x \cap Q_{x'}$ with $x \neq x'$, then (x, q) and (x', q) are two vertices of S in the fibre of q, which is impossible by the previous bullet. Hence $|N_G(p)| \geq |Q_0| + |Q_1| + |Q_2| \geq 6$, contradicting the degree bound.

So every fibre contains exactly one vertex of S, and $S = S_f$ for the map f defined by $(f(p), p) \in S$. This f is unique because S_f determines f. By the computation above, f is a fall 3-colouring. The final statement follows: $f \mapsto S_f$ is injective, it maps fall 3-colourings to 2-d kernels by (a), and it is onto the 2-d kernels by (b). □

7.2 Proper 3-colourings of cycles

Lemma 7.3. *For $k \geq 3$, the number of proper colourings $g : \mathbb{Z}_k \to \mathbb{Z}_3$ of C_k is $\chi(k) = 2^k + 2(-1)^k$.*

Proof. A map g is a proper colouring of C_k if and only if $g(u) \neq g(u+1)$ for every $u \in \mathbb{Z}_k$, that is, if and only if $(g(0), g(1), \ldots, g(k-1))$ is a closed walk of length k in the complete graph K_3 on $\mathbb{Z}_3$ (viewed as a digraph with arcs in both directions). By Fact 2.5 their number is $\operatorname{tr}(B^k)$, where $B = J - I$ and J is the 3×3 all-ones matrix. The vector $\mathbf{1} = (1,1,1)^{\mathsf{T}}$ satisfies $B\mathbf{1} = 3\mathbf{1} - \mathbf{1} = 2\mathbf{1}$. The vectors $v_1 = (1,-1,0)^{\mathsf{T}}$ and $v_2 = (0,1,-1)^{\mathsf{T}}$ have coordinate sum 0, so $Jv_1 = Jv_2 = 0$ and $Bv_1 = -v_1$, $Bv_2 = -v_2$. The three vectors $\mathbf{1}, v_1, v_2$ are linearly independent (the matrix with these columns has determinant $3 \neq 0$), so with $Q = (\mathbf{1}\ v_1\ v_2)$ we have $B = Q \operatorname{diag}(2,-1,-1)\, Q^{-1}$ and $B^k = Q \operatorname{diag}(2^k, (-1)^k, (-1)^k)\, Q^{-1}$. Since the trace is invariant under conjugation, $\operatorname{tr}(B^k) = 2^k + 2(-1)^k$. □

This is the value at 3 of the chromatic polynomial $(\lambda - 1)^k + (-1)^k(\lambda - 1)$ of the cycle; see, for example, [4].

7.3 Fall 3-colourings of two-dimensional tori

Throughout this subsection $b, c \geq 3$, the torus $C_b \,\square\, C_c$ has vertex set $\mathbb{Z}_b \times \mathbb{Z}_c$, and its vertices (y, z) have the four distinct neighbours $(y \pm 1, z)$ (*horizontal*) and $(y, z \pm 1)$ (*vertical*). The set $\{y\} \times \mathbb{Z}_c$ is the *column* of y and $\mathbb{Z}_b \times \{z\}$ the *row* of z. When $3 \mid b$, the reduction $\mathbb{Z}_b \to \mathbb{Z}_3$, $y \mapsto y \bmod 3$, is well defined, and for $\varepsilon \in \{1, -1\}$ we write εy for $\varepsilon \cdot (y \bmod 3) \in \mathbb{Z}_3$; likewise for $z \in \mathbb{Z}_c$ when $3 \mid c$.

Definition 7.4 (steps and turns)**.** Let f be a proper 3-colouring of $C_b \,\square\, C_c$. Its *horizontal* and *vertical steps* are

$$\alpha(y, z) = f(y + 1, z) - f(y, z), \qquad \beta(y, z) = f(y, z + 1) - f(y, z) \qquad (\text{in } \mathbb{Z}_3).$$

By (F1), α and β never take the value 0, so they take values in $\{1, -1\}$. A vertex (y, z) is a *horizontal turn* if $\alpha(y - 1, z) \neq \alpha(y, z)$, and a *vertical turn* if $\beta(y, z - 1) \neq \beta(y, z)$.

Lemma 7.5. *Let f be a proper 3-colouring and (y, z) a vertex. Then (y, z) is a horizontal turn if and only if $f(y - 1, z) = f(y + 1, z)$. In that case $f(y - 1, z) = f(y + 1, z) = f(y, z) + \delta$ with $\delta = \alpha(y, z) \in \{1, -1\}$.*

Proof. By definition of α, $f(y - 1, z) = f(y, z) - \alpha(y - 1, z)$ and $f(y + 1, z) = f(y, z) + \alpha(y, z)$. These are equal if and only if $\alpha(y, z) = -\alpha(y - 1, z)$, which, for values in $\{1, -1\}$, holds if and only if $\alpha(y, z) \neq \alpha(y - 1, z)$. In that case both equal $f(y, z) + \alpha(y, z)$. □

Lemma 7.6 (sufficiency)**.** *(i) If $3 \mid b$, $\varepsilon \in \{1, -1\}$ and g is a proper 3-colouring of C_c, then $f(y, z) = \varepsilon y + g(z)$ is a fall 3-colouring of $C_b \,\square\, C_c$.*

(ii) If $3 \mid c$, $\varepsilon \in \{1, -1\}$ and h is a proper 3-colouring of C_b, then $f(y, z) = h(y) + \varepsilon z$ is a fall 3-colouring of $C_b \,\square\, C_c$.

Proof. (i) The map is well defined because $3 \mid b$. For every vertex, $\alpha(y, z) = \varepsilon(y + 1) - \varepsilon y = \varepsilon \neq 0$ and $\beta(y, z) = g(z + 1) - g(z) \neq 0$ because g is proper; so f is proper. For (F2), the vertex (y, z) of colour k has horizontal neighbours $(y + 1, z)$ and $(y - 1, z)$ of colours $k + \varepsilon$ and $k - \varepsilon$, which are the two colours different from k. (ii) is the same argument with the roles of the coordinates exchanged. □

Lemma 7.7 (a turn spreads along its column)**.** *Let f be a fall 3-colouring of $C_b \,\square\, C_c$ and let (y, z) be a horizontal turn. Then every vertex of the column of y is a horizontal turn, and $\beta(y, \cdot)$ is constant on $\mathbb{Z}_c$.*

Proof. Put $k_0 = f(y, z)$ and let δ be as in Lemma 7.5, so both horizontal neighbours of (y, z) have colour $k_0 + \delta$. The three colours k_0, $k_0 + \delta$, $k_0 - \delta$ are distinct (as $\delta \neq 0$ and $2\delta \neq 0$ in $\mathbb{Z}_3$), so they are all of $\mathbb{Z}_3$. By (F2), (y, z) has a neighbour of colour $k_0 - \delta$; it is not a horizontal neighbour, so there is $\eta \in \{1, -1\}$ with $f(y, z + \eta) = k_0 - \delta$.

For $j \geq 1$ let D_j be the statement

$$f(y, z + j\eta) = k_0 - j\delta, \qquad f(y, z + (j - 1)\eta) = k_0 - (j - 1)\delta, \qquad f(y \pm 1, z + j\eta) = k_0 - (j - 1)\delta.$$

We prove D_j for all $j \geq 1$ by induction.

D_1. The first two equalities are $f(y, z + \eta) = k_0 - \delta$ and $f(y, z) = k_0$. For the third, the vertex $(y \pm 1, z + \eta)$ is adjacent to $(y \pm 1, z)$, of colour $k_0 + \delta$, and to $(y, z + \eta)$, of colour $k_0 - \delta$. By (F1) its colour is the remaining colour k_0.

$D_j \Rightarrow D_{j+1}$. Let $w = (y, z + j\eta)$ and $k_j = k_0 - j\delta = f(w)$. By D_j, both horizontal neighbours of w have colour $k_0 - (j-1)\delta = k_j + \delta$, and its vertical neighbour $(y, z + (j-1)\eta)$ has colour $k_j + \delta$ as well. By (F2), w has a neighbour of colour $k_j - \delta \neq k_j + \delta$. Its only remaining neighbour is $(y, z + (j+1)\eta)$, which is different from $(y, z + (j-1)\eta)$ because $2\eta \not\equiv 0 \pmod{c}$ (as $c \geq 3$). Hence $f(y, z + (j+1)\eta) = k_j - \delta = k_0 - (j+1)\delta$. Next, $(y \pm 1, z + (j+1)\eta)$ is adjacent to $(y \pm 1, z + j\eta)$, of colour $k_j + \delta$, and to $(y, z + (j+1)\eta)$, of colour $k_j - \delta$; by (F1) its colour is $k_j = k_0 - j\delta$. Together with $f(y, z + j\eta) = k_0 - j\delta$ this is D_{j+1}.

Conclusion. For every $j \geq 0$ we have shown $f(y, z + j\eta) = k_0 - j\delta$ and $f(y \pm 1, z + j\eta) = f(y, z + j\eta) + \delta$ (for $j = 0$ this is the hypothesis). By Lemma 7.5, every vertex $(y, z + j\eta)$ is a horizontal turn. As j runs over $0, 1, \ldots, c-1$, $z + j\eta$ runs over all of $\mathbb{Z}_c$. For the vertical steps:

- if $\eta = 1$: $\beta(y, z + j) = f(y, z + j + 1) - f(y, z + j) = -\delta$ for all $j \geq 0$;
- if $\eta = -1$: $\beta(y, z - j - 1) = f(y, z - j) - f(y, z - j - 1) = (k_0 - j\delta) - (k_0 - (j+1)\delta) = \delta$ for all $j \geq 0$.

In both cases $\beta(y, \cdot)$ is constant on $\mathbb{Z}_c$. □

Lemma 7.8 (no turns)**.** *Let f be a proper 3-colouring of $C_b \,\square\, C_c$ with no horizontal turn. Then $3 \mid b$, and there are $\varepsilon \in \{1, -1\}$ and a proper 3-colouring g of C_c with $f(y, z) = \varepsilon y + g(z)$ for all (y, z).*

Proof. Constant step along each row. Fix z. Since no vertex is a horizontal turn, $\alpha(y-1, z) = \alpha(y, z)$ for every $y \in \mathbb{Z}_b$, hence $\alpha(y, z) = \alpha(0, z)$ for all y; call this value $\varepsilon_z \in \{1, -1\}$.

$3 \mid b$. Identify $\mathbb{Z}_b$ with $\{0, 1, \ldots, b-1\}$. By induction on t, $f(t, z) = f(0, z) + t\varepsilon_z$ for $0 \leq t \leq b-1$, and one more step gives $f(0, z) = f(b-1, z) + \varepsilon_z = f(0, z) + b\varepsilon_z$. So $b\varepsilon_z = 0$ in $\mathbb{Z}_3$; multiplying by ε_z (note $\varepsilon_z^2 = 1$) gives $b \equiv 0 \pmod{3}$.

The steps agree across rows. Suppose $\varepsilon_{z+1} \neq \varepsilon_z$ for some z. Then $\lambda = \varepsilon_{z+1} - \varepsilon_z \in \{2, -2\} = \{-1, 1\}$ in $\mathbb{Z}_3$. Put $\kappa = f(0, z+1) - f(0, z)$. For $t = 0, 1, 2$ (vertices, since $b \geq 3$),

$$f(t, z+1) - f(t, z) = \big(f(0, z+1) + t\varepsilon_{z+1}\big) - \big(f(0, z) + t\varepsilon_z\big) = \kappa + t\lambda.$$

The three values κ, $\kappa + \lambda$, $\kappa + 2\lambda$ are pairwise distinct (their differences are λ, 2λ and λ, all nonzero), so they are all of $\mathbb{Z}_3$, and one of them is 0. Then $f(t, z+1) = f(t, z)$ for some t, contradicting (F1) on the vertical edge between (t, z) and $(t, z+1)$. Hence all ε_z are equal; call the common value ε.

The form of f. Put $g(z) = f(0, z)$. For $y \in \mathbb{Z}_b$ with representative $t \in \{0, \ldots, b-1\}$ we have $f(y, z) = g(z) + t\varepsilon = \varepsilon y + g(z)$, where εy is well defined since $3 \mid b$. Finally $g(z+1) - g(z) = \beta(0, z) \neq 0$, so g is a proper colouring of C_c. □

Symmetry. The map $(y, z) \mapsto (z, y)$ is an isomorphism from $C_b \,\square\, C_c$ to $C_c \,\square\, C_b$ that exchanges horizontal and vertical edges, rows and columns, and horizontal and vertical turns, and f is a fall 3-colouring of $C_b \,\square\, C_c$ if and only if f composed with the inverse map is one of $C_c \,\square\, C_b$. Applying Lemmas 7.7 and 7.8 to $C_c \,\square\, C_b$ and translating back gives:

Lemma 7.9 (vertical versions)**.** *Let f be a fall 3-colouring of $C_b \,\square\, C_c$.*

(a) If some vertex of the row of z is a vertical turn, then every vertex of that row is a vertical turn and $\alpha(\cdot, z)$ is constant on $\mathbb{Z}_b$.

(b) If f has no vertical turn, then $3 \mid c$, and there are $\varepsilon \in \{1, -1\}$ and a proper 3-colouring h of C_b with $f(y, z) = h(y) + \varepsilon z$.

Theorem 7.10 (classification of fall 3-colourings of tori)**.** *Let $b, c \geq 3$. A map $f : \mathbb{Z}_b \times \mathbb{Z}_c \to \mathbb{Z}_3$ is a fall 3-colouring of $C_b \,\square\, C_c$ if and only if one of the following holds:*

(i) $3 \mid b$ *and* $f(y,z) = \varepsilon y + g(z)$ *for some* $\varepsilon \in \{1,-1\}$ *and some proper 3-colouring g of C_c;*

(ii) $3 \mid c$ *and* $f(y,z) = h(y) + \varepsilon z$ *for some* $\varepsilon \in \{1,-1\}$ *and some proper 3-colouring h of C_b.*

In particular $C_b \,\square\, C_c$ has a fall 3-colouring if and only if $3 \mid b$ or $3 \mid c$.

Proof. ($\Leftarrow$) Lemma 7.6.

($\Rightarrow$) Let f be a fall 3-colouring.

Case A: f has no horizontal turn. Then (i) holds by Lemma 7.8.

Case B: some vertex (y_0, z_0) is a horizontal turn. By Lemma 7.7, $\beta(y_0, \cdot)$ is constant, so $\beta(y_0, z-1) = \beta(y_0, z)$ for every z, and no vertex of the column of y_0 is a vertical turn. Suppose some vertex (y_1, z_1) were a vertical turn. By Lemma 7.9(a), every vertex of the row of z_1 would be a vertical turn, including (y_0, z_1), which lies in the column of y_0. This is a contradiction, so f has no vertical turn, and (ii) holds by Lemma 7.9(b).

For the last statement: if $3 \mid b$, take $\varepsilon = 1$ and any proper colouring g of C_c (cycles are 3-colourable, for example by Lemma 7.3, since $\chi(c) \geq 2^3 - 2 > 0$), and use (i); if $3 \mid c$ use (ii). Conversely, a fall 3-colouring satisfies (i) or (ii), each of which requires $3 \mid b$ or $3 \mid c$. □

Corollary 7.11 (counting fall 3-colourings)**.** *Let $b, c \geq 3$. The number of fall 3-colourings of $C_b \square C_c$ is*

$$\Phi(b,c) = 2\chi(c)\,[\,3 \mid b\,] + 2\chi(b)\,[\,3 \mid c\,] - 12\,[\,3 \mid b \text{ and } 3 \mid c\,], \qquad \chi(k) = 2^k + 2(-1)^k.$$

Proof. Let $\mathcal{A}$ and $\mathcal{B}$ be the sets of maps of the forms (i) and (ii) of Theorem 7.10; we take $\mathcal{A} = \emptyset$ if $3 \nmid b$ and $\mathcal{B} = \emptyset$ if $3 \nmid c$. By the theorem, the number of fall 3-colourings is $|\mathcal{A} \cup \mathcal{B}| = |\mathcal{A}| + |\mathcal{B}| - |\mathcal{A} \cap \mathcal{B}|$.

$|\mathcal{A}|$. Let $3 \mid b$. The pair (ε, g) is recovered from $f(y,z) = \varepsilon y + g(z)$ by $\varepsilon = f(1,0) - f(0,0)$ and $g(z) = f(0,z)$, so distinct pairs give distinct maps. There are 2 choices of ε and $\chi(c)$ choices of g (Lemma 7.3), so $|\mathcal{A}| = 2\chi(c)$.

$|\mathcal{B}|$. Symmetrically, $|\mathcal{B}| = 2\chi(b)$ when $3 \mid c$.

$|\mathcal{A} \cap \mathcal{B}|$. This is empty unless $3 \mid b$ and $3 \mid c$, so assume both. If $f \in \mathcal{A} \cap \mathcal{B}$, then from (i) the horizontal step of f is the constant ε, and from (ii) the vertical step is a constant ε'; hence $f(y,z) = f(0,0) + \varepsilon y + \varepsilon' z$. Conversely every map of this form lies in $\mathcal{A}$ (with $g(z) = f(0,0) + \varepsilon' z$, a proper colouring of C_c because its steps are $\varepsilon' \neq 0$ and it is well defined as $3 \mid c$) and in $\mathcal{B}$ (symmetrically). The triple $(f(0,0), \varepsilon, \varepsilon')$ is recovered from f, so there are $3 \cdot 2 \cdot 2 = 12$ such maps.

Adding up gives $\Phi(b,c)$. □

7.4 The three-dimensional result

Corollary 7.12. *Let $b, c \geq 3$. The 2-d kernels of $C_3 \,\square\, C_b \,\square\, C_c$ are exactly the sets $\{(f(y,z), y, z) : (y,z) \in \mathbb{Z}_b \times \mathbb{Z}_c\}$ with f a fall 3-colouring of $C_b \,\square\, C_c$, and*

$$\sigma(C_3 \,\square\, C_b \,\square\, C_c) = \Phi(b,c) = 2\chi(c)\,[\,3 \mid b\,] + 2\chi(b)\,[\,3 \mid c\,] - 12\,[\,3 \mid b \text{ and } 3 \mid c\,].$$

In particular $C_3 \,\square\, C_b \,\square\, C_c$ has a 2-d kernel if and only if $3 \mid b$ or $3 \mid c$.

Proof. By Definition 1.3, $C_3 \,\square\, C_b \,\square\, C_c = C_3 \,\square\, G$ with $G = C_b \,\square\, C_c$. Every vertex of G has exactly $4 \leq 5$ neighbours. Lemma 7.2 identifies the 2-d kernels with the sets S_f for fall 3-colourings f of G, and Corollary 7.11 counts them. Since $\chi(k) \geq 2^3 - 2 = 6 > 0$ for $k \geq 3$, the count is positive if and only if $3 \mid b$ or $3 \mid c$: if exactly one holds it is $2\chi(\cdot) > 0$, and if both hold it is $2\chi(c) + 2\chi(b) - 12 \geq 12 + 12 - 12 > 0$. □

Example 7.13. Using $\chi(3) = 6$, $\chi(4) = 18$, $\chi(6) = 66$ and $\chi(7) = 126$:

- $\sigma(C_3 \,\square\, C_3 \,\square\, C_3) = 2 \cdot 6 + 2 \cdot 6 - 12 = 12$;
- $\sigma(C_3 \,\square\, C_3 \,\square\, C_4) = 2\chi(4) = 36$;
- $\sigma(C_3 \,\square\, C_6 \,\square\, C_6) = 2 \cdot 66 + 2 \cdot 66 - 12 = 252$;
- $\sigma(C_3 \,\square\, C_7 \,\square\, C_7) = 0$ and $\sigma(C_3 \,\square\, C_4 \,\square\, C_4) = 0$.

Remark 7.14. In two dimensions, $C_m \,\square\, C_n$ has a 2-d kernel exactly when the two lengths share a factor (Corollary 6.1). Example 7.13 shows that no condition on pairwise common factors can decide existence in three dimensions: in $C_3 \,\square\, C_7 \,\square\, C_7$ and $C_3 \,\square\, C_4 \,\square\, C_4$ the last two lengths share a factor, yet there is no 2-d kernel.

8 The published statements for tori

In [1] the symbol $C_n \times C_m$ denotes the Cartesian product (its definition is given in the introduction of [1]). Corollary 6.1 contradicts [1, Theorem 2.5 and Corollary 2.6] as follows.

- *Existence.* $C_3 \,\square\, C_6$, $C_3 \,\square\, C_9$, $C_6 \,\square\, C_9$ and $C_5 \,\square\, C_{10}$ have 2-d kernels, since their greatest common divisors are 3, 3, 3 and 5. In each of them the two lengths are neither both even nor equal, so [1, Theorem 2.5] asserts that no 2-d kernel exists. Example 6.2 exhibits one explicitly.
- *Number, both even.* $\sigma(C_6 \,\square\, C_6) = 2P(6) - 2 = 8$ and $\sigma(C_8 \,\square\, C_8) = 2P(8) - 2 = 18$, not 2. For m, n even the value 2 is correct exactly when $P(d) = 2$, that is, when $d \in \{2, 4\}$.
- *Number, odd squares.* $\sigma(C_n \,\square\, C_n) = 2P(n)$ for odd n. This equals $2n$ for $n = 3, 5, 7$, but $\sigma(C_9 \,\square\, C_9) = 2 \cdot 12 = 24 \neq 18$ and $\sigma(C_{11} \,\square\, C_{11}) = 2 \cdot 22 = 44 \neq 22$.

Two steps of the proof of [1, Theorem 2.5] are not justified. For m even and n odd, it assumes that the set chosen in each copy of C_m is a 2-d kernel of C_m; by Lemma 4.2 (and its quotient form) it only has to be a maximal independent set of C_m. For $m \neq n$ both odd, it assumes that the set chosen in each copy of C_m has exactly $\lfloor m/2 \rfloor$ elements, whereas Lemma 3.10 allows every size from $\lceil m/3 \rceil$ to $\lfloor m/2 \rfloor$.

9 Computational verification

The proofs above do not depend on computation. As an independent check, the following scripts in the repository, under `odd-torus/ws-W3/`, compare the formulas with exhaustive enumeration.

- `sat_check.py`: SAT existence for $3 \leq m \leq 14$ and $3 \leq n \leq 40$ (456 tori) agrees with $\gcd(m, n) > 1$; SAT enumeration of all 2-d kernels for $m \leq 12$ and $n \leq 16$ agrees with Corollary 6.1(c), and every kernel found is diagonal; a SAT-free brute force agrees for all $mn \leq 64$.
- `coordinator_check.py`: an independent SAT encoding enumerating all 2-d kernels for $3 \leq m \leq 10$ and $m \leq n \leq 12$; no disagreement with Corollary 6.1(c).
- `transfer.py`: the column transfer digraph for $3 \leq m \leq 20$; the set of n with a 2-d kernel is exactly $\{n \geq 3 : \gcd(m, n) > 1\}$.

- `min_size.py` and `gamma2_sat.py`: minimum sizes for $m, n \leq 12$ agree with Corollary 6.1(d).
- `grid_quotient_check.py`: SAT enumeration for the tensor products $C_m \times C_n$ with $3 \leq m \leq n$ and $mn \leq 64$ (49 graphs), and for all circulants satisfying the hypotheses of Corollary 6.5 with $N \leq 30$ (798 graphs); no disagreement with Corollaries 6.3 and 6.5.
- `threed_check.py`: SAT enumeration of all 2-d kernels of $C_3 \,\square\, C_b \,\square\, C_c$ for $3 \leq b \leq c \leq 9$ (28 graphs); the counts agree with Corollary 7.12, and every kernel meets every triangle fibre in exactly one vertex.
- `fall3.py` and `fall3_rigidity.py`: fall 3-colourability of $C_b \,\square\, C_c$ for $b, c \leq 24$ agrees with Theorem 7.10, and in 13 tori every fall 3-colouring has a constant step along one axis, with the counts of Corollary 7.11.

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